

HRMS Policy Management & Digital Signature

Developer Specification

Policy Documents • Versioning • Approval • Distribution • Acknowledgement • E-Signature • Compliance • Rules

1. Purpose

Build a centralized HRMS Policy Management module where HR can create different policies, upload PDF/images/documents, version them, define applicability, run approvals, publish to employees, collect acknowledgement or electronic signatures, track compliance, and maintain a complete audit trail.

2. Core Principles

- Do not build this as a simple file-upload page.
- Keep human-readable policy documents separate from machine-readable rules, but link them by version.
- Published versions are immutable; changes create a new version.
- Every acknowledgement/signature references the exact policy version.
- Applicability and compliance are calculated server-side.
- Verify e-signature completion from the signing provider.
- All actions are tenant-scoped and auditable.

3. Policy Categories

- Leave
- Attendance
- Overtime
- Shift
- Work From Home/Remote Work
- Code of Conduct
- IT/Security
- Travel
- Expense
- Payroll
- Holiday
- Grievance
- Workplace Conduct/POSH
- Data Privacy
- Employee Handbook
- Custom

4. Policy Master

Field	Purpose
policy_id	Unique policy ID
organization_id	Tenant/organization scope
policy_code	Unique business code

policy_name	Policy title
category_id	Category
description	Summary
owner_id	Policy owner
current_version_id	Current version
status	DRAFT/PENDING_APPROVAL/APPROVED/PUBLISHED/ARCHIVED/EXPIRED
created_by	Creator
created_at/updated_at	Audit timestamps

5. Policy Versioning

Field	Purpose
policy_version_id	Unique version
policy_id	Parent policy
version	e.g. 1.0/2.0
content	Human-readable content
effective_from	Effective date/time
review_date	Review date
expiry_date	Optional expiry
signature_mode	NONE/ACKNOWLEDGEMENT/E_SIGNATURE/BOTH
due_date	Employee deadline
status	Version lifecycle state

Never edit a published version in place. Historical versions must remain available for audit.

6. PDF / Image / Document Upload

- Support PDF, DOC/DOCX, JPG, JPEG, PNG, with additional safe formats configurable.
- Allow one main document and multiple supporting attachments per policy version.
- Store filename, MIME type, size, checksum/hash, storage reference, uploader and upload time.
- Use private/secured storage; do not expose unrestricted public URLs.
- Validate files server-side and malware-scan where available.
- Do not silently overwrite attachments belonging to a published version.

7. Policy Applicability

Scope	Example
Organization	All employees
Country/Location	India/UK/USA
Legal Entity	Specific entity
Department	IT/HR/Finance
Designation/Level	Developer/Manager
Employment Type	Full-time/Contract/Intern
Employee Group	Custom group
Individual	Specific employee(s)

Support configurable AND/OR rules such as India AND IT, or Managers OR Directors.

8. Approval Workflow

Recommended lifecycle: DRAFT → PENDING_APPROVAL → APPROVED → SCHEDULED/PUBLISHED → ACTIVE → SUPERSEDED/ARCHIVED/EXPIRED.

1. Create/upload policy.

2. Submit version for approval.
3. Approvers review document, attachments and rules.
4. Approve, reject or request changes.
5. Publish immediately or schedule publication.
6. Generate compliance tasks only for applicable employees.

9. Approval Matrix

- HR only
- HR + Department Head
- HR + Legal/Compliance
- HR + Management
- Custom multi-level approval

Every approval action stores approver, role/level, action, comment and timestamp.

10. Employee Policy Center

- My Policies
- Mandatory Policies
- Pending acknowledgement
- Pending signature
- Completed policies
- Policy version/effective date
- Deadline
- Attachments
- View/Download
- Acknowledgement/signature history

11. Acknowledgement Modes

Mode	Behavior
NONE	Informational only
ACKNOWLEDGEMENT	Employee confirms read/understood
E_SIGNATURE	Electronic signing workflow
BOTH	Acknowledgement + e-signature

An 'I Agree' checkbox must not be represented as a legally valid electronic/digital signature unless the applicable legal/compliance framework explicitly permits it.

12. Electronic / Digital Signature

For legally meaningful signing, integrate an appropriate electronic-signature/e-sign provider for the organization's jurisdiction. HRMS should orchestrate the workflow and retain provider evidence rather than casually implementing its own legal signature mechanism.

7. Employee opens the exact assigned policy version.
8. Employee starts signing.
9. HRMS creates provider transaction/envelope.
10. Provider authenticates signer using configured method.
11. Provider completes signing.

12. Provider sends secure callback/webhook.
13. HRMS validates callback/provider status.
14. HRMS stores signed-document/evidence reference and marks the compliance record SIGNED.

13. Signature Record

Field	Purpose
signature_id	HRMS signature ID
employee_id	Signer
policy_version_id	Exact signed version
provider	E-sign provider
provider_transaction_id	External reference
status	PENDING/SIGNED/DECLINED/EXPIRED/VOID
initiated_at	Start time
signed_at	Completion time
authentication_method	Provider-reported method
document_hash	Integrity reference if supported
signed_document_ref	Stored/provider reference
evidence_ref	Certificate/audit evidence reference if available

14. Distribution & Compliance

Status	Meaning
ASSIGNED	Task created
VIEWED	Employee opened policy
ACKNOWLEDGED	Acknowledgement complete
SIGNATURE_PENDING	Signature required
SIGNED	Signature complete
OVERDUE	Deadline passed
DECLINED	Declined where applicable

Dashboard should show total assigned, viewed, acknowledged, signed, pending, overdue and declined.

15. Reminders & Escalation

- Initial notification
- Configurable reminders
- Deadline reminder
- Manager escalation
- HR escalation
- Overdue status
- Optional final compliance action

Frequency, timing and recipients must be configurable.

16. Policy Rules Engine

Policies that drive HRMS behavior should have both a human-readable document and structured rules.

Policy	Structured rule examples
Attendance	Grace period, late threshold, rounding
Overtime	Minimum OT, approval, holiday/week-off rules, limits
Leave	Accrual, carry-forward, eligibility

Shift	Break rules, night-shift rules
Payroll	Eligibility/cutoff/approval rules

Rules must be versioned/effective-dated so historical calculations remain reproducible.

17. HRMS Module Integration

Policy Management → Policy Rules → Attendance / Leave / Shift / OT / Payroll.

- Attendance resolves applicable Attendance Policy version.
- OT resolves applicable OT Policy version.
- Leave resolves applicable Leave Policy version.
- Payroll retains the policy/rule version used for calculation.

18. Policy Package

A policy version may contain Main Policy.pdf plus supporting PDFs/images/documents. Present these as one package. Where required, create a consolidated download/signing package without modifying source documents.

19. Suggested Database Tables

Table	Important fields
policies	id, organization_id, code, name, category_id, owner_id, current_version_id, status
policy_versions	id, policy_id, version, content, effective_from, review_date, expiry_date, signature_mode, due_date, status
policy_attachments	id, policy_version_id, filename, mime_type, size, checksum, storage_ref, uploaded_by
policy_applicability_rules	id, policy_version_id, scope_type, scope_id, operator, rule_group
policy_approvals	id, policy_version_id, approver_id, level, action, comment, action_at
policy_distributions	id, policy_version_id, employee_id, assigned_at, due_at, status, viewed_at, acknowledged_at
policy_acknowledgements	id, distribution_id, employee_id, policy_version_id, status, acknowledged_at
policy_signatures	id, distribution_id, employee_id, policy_version_id, provider, provider_transaction_id, status, signed_at, document_hash, evidence_ref
policy_reminders	id, distribution_id, reminder_type, scheduled_at, sent_at, status
policy_rules	id, policy_version_id, rule_type, rule_json, effective_from, status
policy_audit_logs	id, organization_id, entity_type, entity_id, action, actor_id, before_json, after_json, created_at

20. Suggested APIs

Method / Endpoint	Purpose
POST /api/policies	Create policy
GET /api/policies	Search/list policies
GET /api/policies/{id}	Policy details
POST /api/policies/{id}/versions	Create version

POST /api/policies/{id}/attachments	Upload attachment
POST /api/policies/versions/{id}/submit	Submit for approval
POST /api/policies/versions/{id}/approve	Approve
POST /api/policies/versions/{id}/reject	Reject/request changes
POST /api/policies/versions/{id}/publish	Publish
GET /api/me/policies	Employee policy center
POST /api/policy-distributions/{id}/acknowledge	Acknowledge
POST /api/policy-distributions/{id}/sign	Start e-sign
POST /api/integrations/esign/webhook	Provider callback
GET /api/policies/compliance	Compliance dashboard
GET /api/policies/{id}/audit	Audit trail

21. Role-Based Access

Role	Typical permissions
HR Admin	Create/edit/upload/configure/approve/publish/compliance
Policy Owner	Create/edit assigned policies/submit
Approver	Review/approve/reject assigned versions
Employee	View applicable policies/acknowledge/sign
Manager	Team compliance/escalations
Legal/Compliance	Review assigned policies
System Admin	Integration/storage administration subject to content-access policy

22. Audit Trail

- Created
- Edited
- Version created
- Attachment uploaded
- Submitted
- Approved
- Rejected
- Published
- Viewed
- Downloaded
- Acknowledged
- Signature initiated
- Signed
- Declined
- Expired
- Archived
- Revoked

Each audit record must include actor, organization, timestamp, entity/version, action, and relevant before/after values or references.

23. Security & Privacy

- Tenant isolation on every policy/document/compliance query.
- Private document storage and short-lived access links where appropriate.

- Server-side authorization and applicability evaluation.
- Secure e-sign webhook verification and replay protection.
- Encrypt sensitive data in transit and at rest.
- Apply configurable retention/legal-hold rules.
- Collect only necessary signer/compliance data.

24. Policy Lifecycle / Historical Behavior

Example: Attendance Policy v3.0 is active. HR creates v4.0 with a changed grace period. v3.0 remains immutable. From v4.0 effective date onward, new attendance calculations use v4.0. Historical finalized records are not silently changed.

25. Example — Attendance Policy

Upload Attendance Policy.pdf + Shift Rules.pdf + diagram.png → create v3.0 → applicability India/All Employees → BOTH acknowledgement and e-signature → approval → publish → employees receive v3.0 → Attendance engine consumes structured v3.0 rules → compliance is stored against v3.0.

26. Example — OT Policy

OT Policy v2.0 contains PDF plus structured rules such as minimum OT, approval requirement and holiday/week-off treatment. OT engine calculates Potential OT from v2.0; approval produces Approved OT; payroll consumes approved amount and retains policy/rule reference.

27. Background Jobs

- Scheduled publication
- Distribution generation
- Reminder scheduler
- Overdue escalation
- Policy review/expiry reminders
- E-sign webhook reconciliation
- Compliance reconciliation
- Archive/retention jobs

28. Testing Requirements

Area	Tests
CRUD	Create/edit/archive
Files	PDF/image/doc, invalid type, size limit
Versioning	Published version immutable
Applicability	Location/department/employment combinations
Approval	Approve/reject/multi-level
Publishing	Immediate/scheduled
Distribution	Only applicable employees
Acknowledgement	Pending → acknowledged
E-sign	Transaction → callback → signed
Webhook	Invalid/duplicate/replayed callback
Reminder	Due/overdue/escalation
Rules	Correct effective version
Audit	All significant actions

Security	Tenant isolation
Retention	Archive/expiry

29. Acceptance Criteria

- HR can create multiple/custom policy categories.
- Each policy version supports multiple attachments.
- Published versions are immutable.
- Applicability rules target only intended employees.
- Approval workflow is configurable.
- Policies can publish now or later.
- Employee Policy Center shows applicable policies and compliance status.
- Acknowledgement and e-signature are separate concepts.
- E-sign completion is verified through provider status/callback.
- Exact policy version is stored with each acknowledgement/signature.
- Reminders/escalations are configurable.
- Compliance dashboard is accurate.
- Human-readable documents can be paired with structured rules.
- Attendance/Leave/OT/Payroll can resolve the correct effective policy version.
- Complete audit trail is retained.
- Documents are securely stored and tenant-isolated.

30. Developer Must Not Do

- Do not overwrite published versions.
- Do not treat a checkbox as a legally valid signature by default.
- Do not store files in unrestricted public storage.
- Do not let frontend determine applicability or signature status.
- Do not let employees sign an unintended version.
- Do not silently alter historical compliance records.
- Do not calculate HR rules from unversioned configuration.
- Do not trust browser-only e-sign success.
- Do not allow cross-organization access.

31. Final Architecture

Policy Master → Version → Documents + Structured Rules → Applicability → Approval → Publish → Employee Distribution
→ Acknowledgement/E-Signature → Compliance → Audit → Attendance/Leave/OT/Payroll Rule Engines